WORKED SOLUTION

# STEP I 1988, Question 15

Probability and Statistics • Binomial Hypothesis Test

**STEP I 1988, Question 15**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q15

## The key idea

Turn "more than half its points from fouls" into a simple inequality between the number of fouls and the number of goals. Conditioning on the number of goals then leaves a count of equally likely foul values. For the second part, ask how many nine-foul occasions the *old* distribution would lead us to expect in 300 tries, and how surprising the observed number would be if nothing had changed.

## Solution

### The probability

For a particular team in a game, let $G$ be the number of goals it scores and $F$ the number of fouls committed against it. Then $G$ and $F$ are independent, with

$$\mathrm{P}(G=0),\ \ldots,\ \mathrm{P}(G=4) \;=\; \tfrac{1}{9},\ \tfrac{2}{9},\ \tfrac{3}{9},\ \tfrac{2}{9},\ \tfrac{1}{9}, \qquad \mathrm{P}(F=f)=\tfrac{1}{10} \quad (f=0,1,\ldots,9).$$

The team scores $2G$ points from goals and $F$ from fouls, so $2G+F$ in total. It gains more than half its points from fouls exactly when

$$F > \tfrac{1}{2}(2G+F) \iff F > 2G.$$

(If the team scores no points at all, then $F=0=2G$ and it does not count, which the strict inequality handles correctly.)

Given $G=g$, the favourable values of $F$ are $2g+1,\ldots,9$, which is $9-2g$ of the ten equally likely values. So, using the independence of $F$ and $G$,

$$\mathrm{P}(F>2G \mid G=g) = \frac{9-2g}{10}, \qquad g=0,1,2,3,4,$$

giving $\frac{9}{10}, \frac{7}{10}, \frac{5}{10}, \frac{3}{10}, \frac{1}{10}$. By the law of total probability,

$$\mathrm{P}(F>2G) = \frac{1\cdot 9+2\cdot 7+3\cdot 5+2\cdot 3+1\cdot 1}{9\times 10} = \frac{45}{90} = \frac{1}{2}.$$

The neat answer has a reason: the goal distribution is symmetric about 2, and $(9-2g)/10$ is linear in $g$, so its average equals its value at $g=2$, namely $\frac{1}{2}$.

#### Has the foul distribution changed?

The number of fouls a team commits is the number committed against its opponents, so under the old rules each team commits 9 fouls in a game with probability $\frac{1}{10}$. Let $X$ be the number of the 300 team-games in which a team committed 9 fouls. We test

$$H_0:\ p=\tfrac{1}{10}\quad\text{(no change)}\qquad\text{against}\qquad H_1:\ p<\tfrac{1}{10},$$

where $p$ is the probability of a team committing 9 fouls in a game. The alternative is one-sided because the rule change was meant to make large numbers of fouls *less* likely.

**?** **Ask yourself** Is a binomial model justified here?

We need the 300 team-games to be independent, each with the same probability $p$. The question gives independence within a game, and it is reasonable to assume that different games are independent too. So under $H_0$, $X\sim \mathrm{B}(300,\frac{1}{10})$, with

$$\mathrm{E}(X)=30,\qquad \mathrm{Var}(X)=300\times\tfrac{1}{10}\times\tfrac{9}{10}=27,\qquad \text{standard deviation } \sqrt{27}\approx 5.20.$$

We observed $X=12$, which is 18 below the expected 30, about 3.5 standard deviations. With the normal approximation and a continuity correction,

$$\mathrm{P}(X\le 12)\approx\Phi\left(\frac{12.5-30}{\sqrt{27}}\right)=\Phi(-3.37)\approx 0.0004.$$

The exact binomial probability is even smaller, about 0.0001. Either way, a result this extreme would happen less than once in a thousand seasons if nothing had changed. It is far beyond the usual 5% or even 1% significance levels, so we reject $H_0$. This *is* good evidence that the distribution of the number of fouls has changed, and in the hoped-for direction: nine-foul games have become rarer.

Two caveats are worth stating. The test only shows that $\mathrm{P}(9\text{ fouls})$ has gone down, not how the rest of the distribution has changed (although that is the natural explanation). Statistics also cannot prove that the rule change *caused* the decrease, since other things may have changed between seasons.

## Checks

- The goal probabilities $\frac{1}{9}(1+2+3+2+1)$ sum to 1, as they must.
- Enumerating all $5\times 10$ pairs $(g,f)$ with exact fractions gives $\mathrm{P}(F>2G)=\frac{1}{2}$, and a simulation of 200,000 games gave 0.4997.
- Summing the binomial probabilities directly gives $\mathrm{P}(X\le 12)=9.2\times10^{-5}$ for $X\sim\mathrm{B}(300,0.1)$. The normal approximation ($3.8\times10^{-4}$) and the Poisson approximation with mean 30 ($1.7\times10^{-4}$) agree that it is tiny.

## Common slips

- Comparing $F$ with $G$ instead of with $2G$: a goal is worth two points.
- Counting $F \geq 2G$ instead of $F > 2G$. "More than half" is strict, and a team with no points gains none of them from fouls.
- Using the 150 games as the sample size instead of the 300 team-games, or forgetting to state the null hypothesis and the binomial model before computing.

## Beyond STEP

The second part is a hypothesis test for a binomial proportion, and the tail probability we found is a $p$-value. At university you'll meet the likelihood ratio and chi-squared tests, which compare a whole observed distribution of foul counts with the old one, not just one value. Blitzstein and Hwang, *Introduction to Probability*, covers the probability side (conditioning, binomial and normal approximations) very clearly.

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